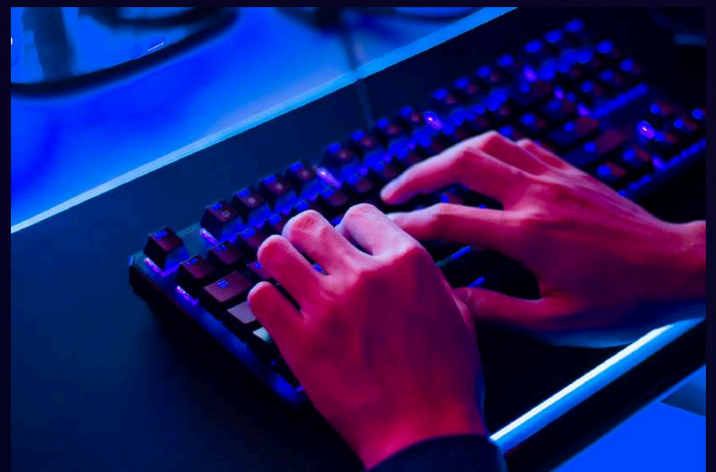
THE ROLE OF C IN SPACE AND DEFENSE SYSTEMS

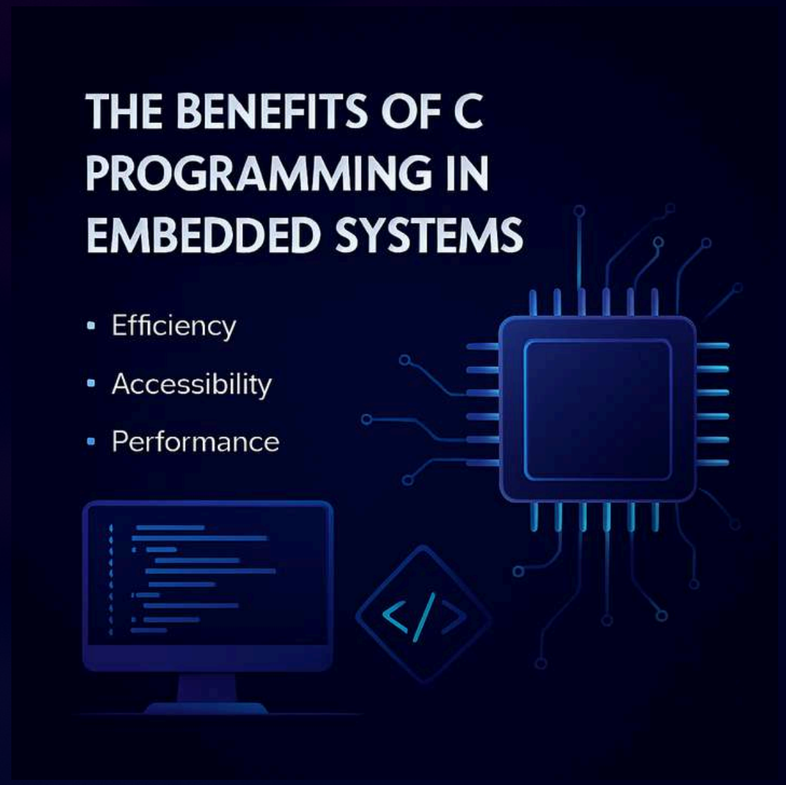
MISSION CRITICAL CODE

Presented by: Aarushi, Devanshi, Lydia, Mrunal, Sahithi

INTRODUCTION TO C IN SPACE AND DEFENSE SYSTEMS

- C programming serves as a foundational language for a multitude of applications, particularly in space and defence systems.
- Its **efficiency, control over system resources, and portability** make it the preferred choice for developing embedded systems that require reliability and performance.





KEY APPLICATION OF C IN SPACE SYSTEMS

01 Satelite Software Development

C programming is essential for the development of software that manages satellite operations, including telemetry, command processing, and data analysis. It ensures reliable communication between the satellite and ground stations.

02 Flight Control Systems

Flight control systems in spacecraft require precise and real-time processing, which C programming provides. It allows for the execution of critical commands that control spacecraft maneuvers and stability during flight.

03 Simulation and Modelling

C programming is widely used in simulation and modeling tools that support mission planning and testing scenarios. These simulations help to predict system behavior and outcomes before actual deployment.

C PROGRAMMING IN DEFENSE SYSTEMS

PROGRAMMING

Weapon Systems Development

C programming plays a crucial role in developing software for precision-guided munitions, which require high accuracy and reliability TLC mission success.

Secure Communication Protocols

In communication systems, C programming enables the creation of secure and efficient protocols, essential for military operations and data integrity during transmission.

Surveillance Systems Software

C is widely used in surveillance systems, particularly in software controlling drones and reconnaissance systems that gather intelligence and monitor activities.

CHALLENGES IN C FOR SPACE AND DEFENSE

01

Safety and Security

In space and defense applications, software must operate flawlessly to prevent failures that could lead to catastrophic outcomes. Ensuring robust safety and security measures is critical to protect against vulnerabilities that could compromise system integrity.

02

Real-time Constraints

Real-time systems require immediate responses to external events. C programming must be optimized to meet stringent timing requirements, ensuring that responses are executed within microseconds to maintain system functionality.

03

Resource Limitations

Space and defense systems often operate on limited hardware resources. C programming must be efficient in memory management and processing power to ensure functionality in environments where resources are constrained.

FUTURE TRENDS IN C PROGRAMMING FOR SPACE AND DEFENCE

2026-2030

Greater use of C in autonomous satellites, drones, and missile systems with improved real-time processing and AI integration

2030-2040

Growth of nano-satellites and compact defense hardware, increasing demand for low-memory, high-efficiency C-based embedded systems.

2040 & BEYOND

Adoption of high-security, safety-certified C standards (MISRA C, secure C) for cyber-resilient and mission-critical space operations.

CONCLUSION

C programming is integral to the advancement and functionality of space and defence systems. Its efficiency and reliability empower developers to create robust applications that are crucial for mission success and operational security.

THANK YOU